



BobSim

TransientEval Lateral Transient Response

Comprehensive Open-Loop Characterization

Generated: 2026-05-12 08:18

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-
- One-period sinusoidal input
 - Continuous sinusoidal response
 - Frequency response computed from fixed-amplitude, single-direction sine sweep
 - Steering shown in degrees for readability

TransientEval Metrics Summary

Time Domain — Step Response

STEP RESPONSE

Peak a_y	5.58	m/s ²
Steady-State a_y	5.10	m/s ²
Steady-State Gain	58.48	(m/s ²)/rad
Overshoot (a_y)	9.3	%
Rise Time (10–90%)	0.11	s
Settling Time	1.23	s
Yaw Overshoot	14.1	%

TransientEval Metrics Summary

Frequency Domain — Sustained Sine

FREQUENCY RESPONSE — CORE

DC Gain (a_y/δ_H)	55.42	(m/s ²)/rad
DC Gain (r/δ_H)	2.77	(rad/s)/rad
Peak Gain (a_y)	55.79	(m/s ²)/rad
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	2.00	Hz
Phase @ 1 Hz (a_y)	-13.6	deg
Phase @ 1 Hz (r)	-10.5	deg
Lag @ 1 Hz (a_y)	0.038	s
Lag @ 1 Hz (r)	0.029	s

DYNAMIC CHARACTER

Gain Slope (a_y)	-0.04	dB/dec
Gain Slope (r)	0.51	dB/dec
Phase Slope (a_y)	41.02	deg/Hz
Phase Slope (r)	26.89	deg/Hz
Phase = -45° (a_y)	2.00	Hz
Phase = -45° (r)	2.00	Hz

RESPONSE COUPLING

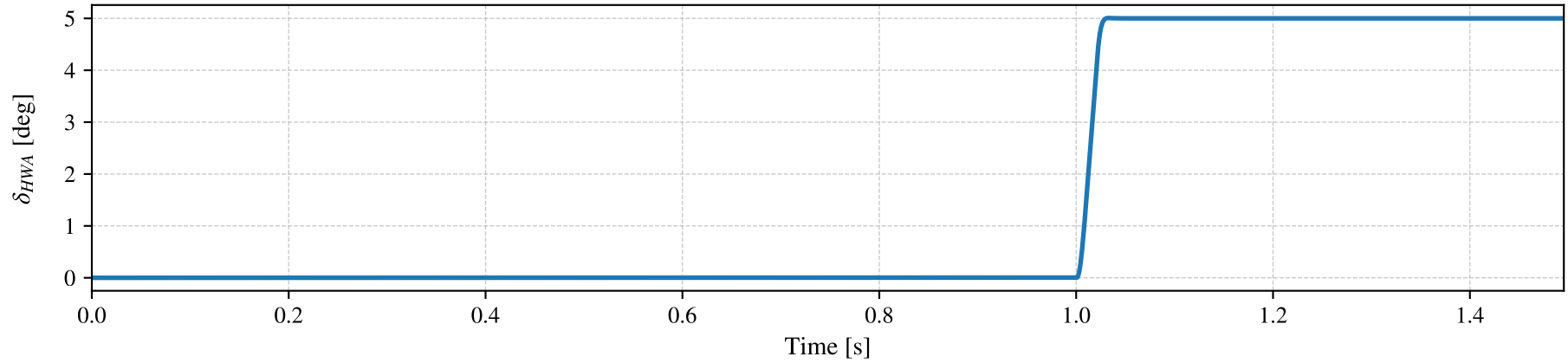
Lag: $\delta_H \rightarrow a_y$	0.038	s
Lag: $\delta_H \rightarrow r$	0.029	s
Lag: $r \rightarrow a_y$	0.009	s
r/a_y Ratio	0.050	(rad/s)/(m/s ²)

QUALITY / VALIDITY

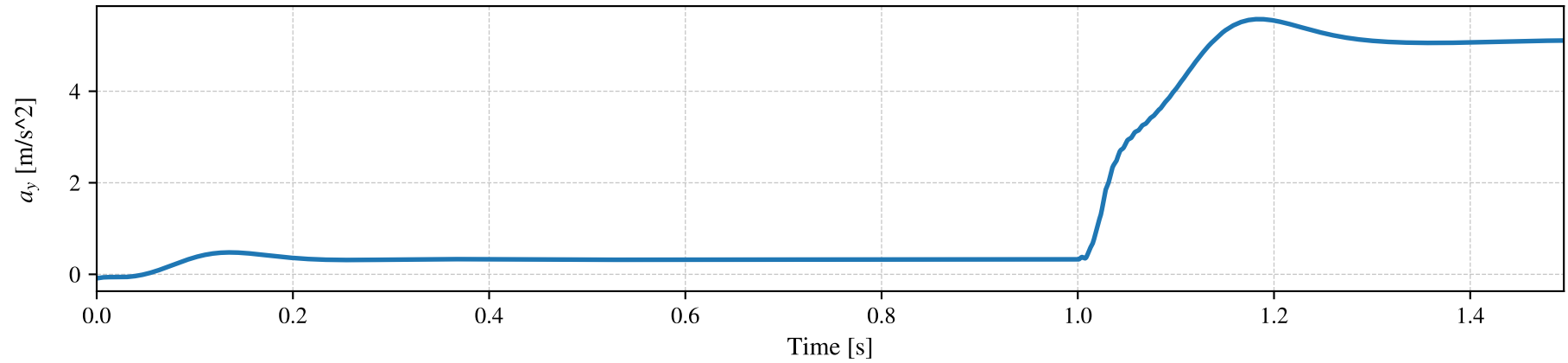
a_y Fit Error	1.15e-02	-
Yaw Fit Error	4.17e-03	-
Gain Variation	1.2	%

Step Steer Response (TransientEval)

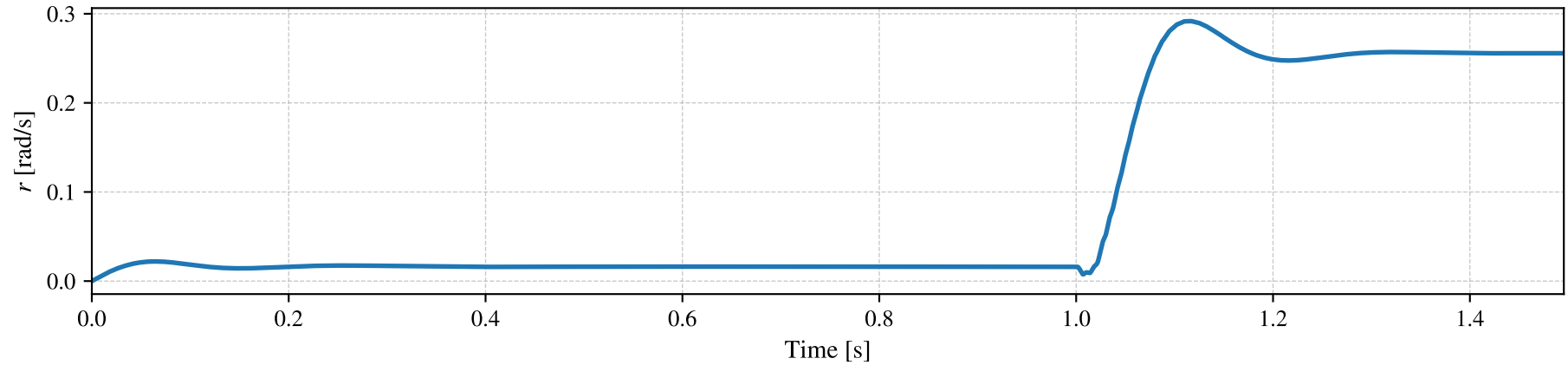
Steering Wheel Angle δ_H



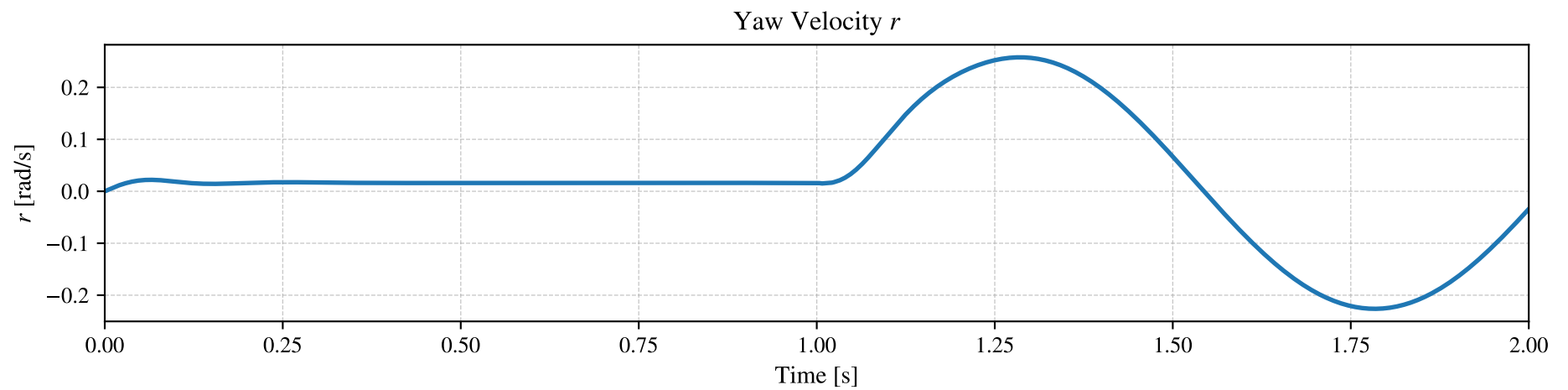
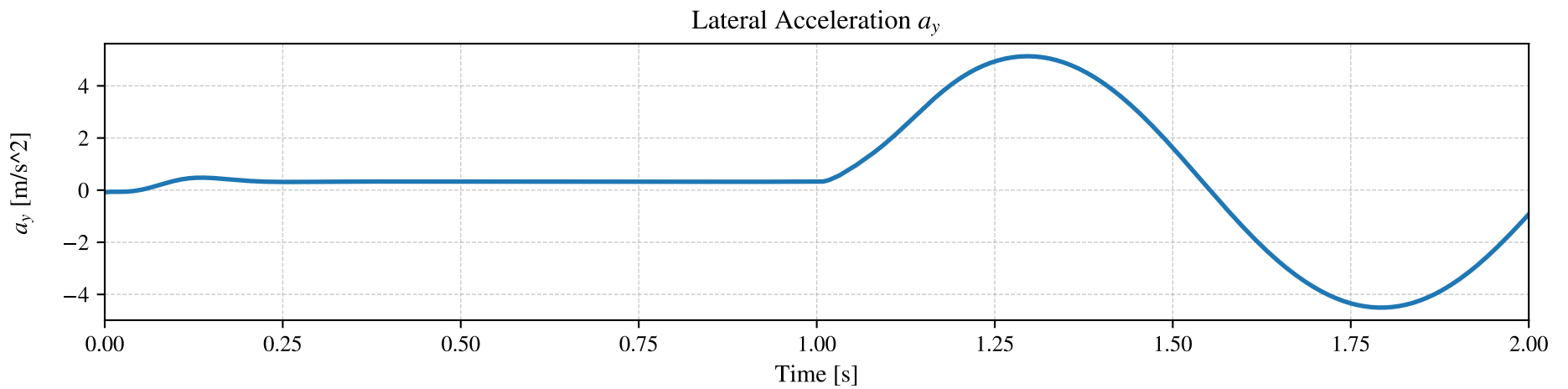
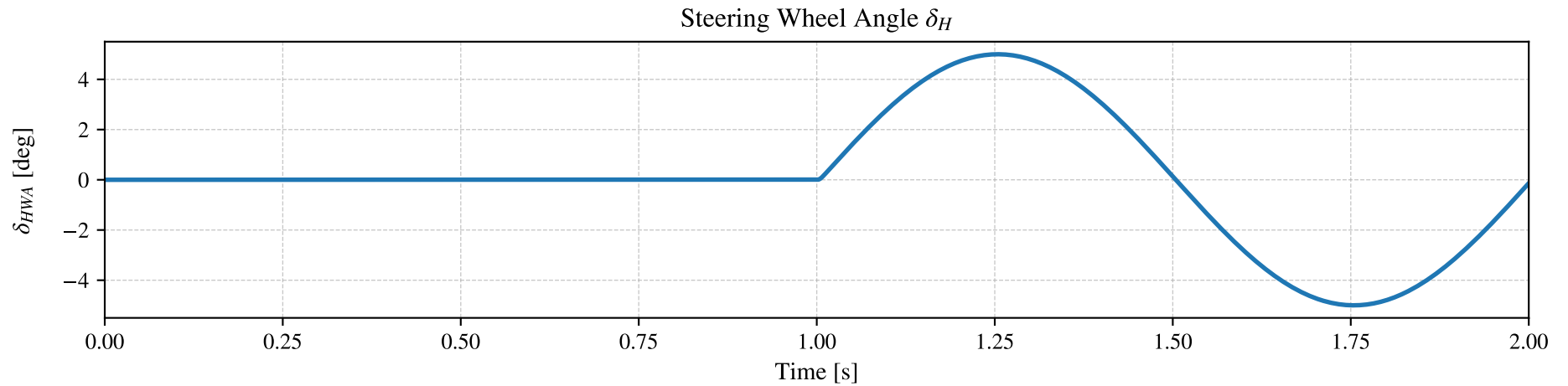
Lateral Acceleration a_y



Yaw Velocity r

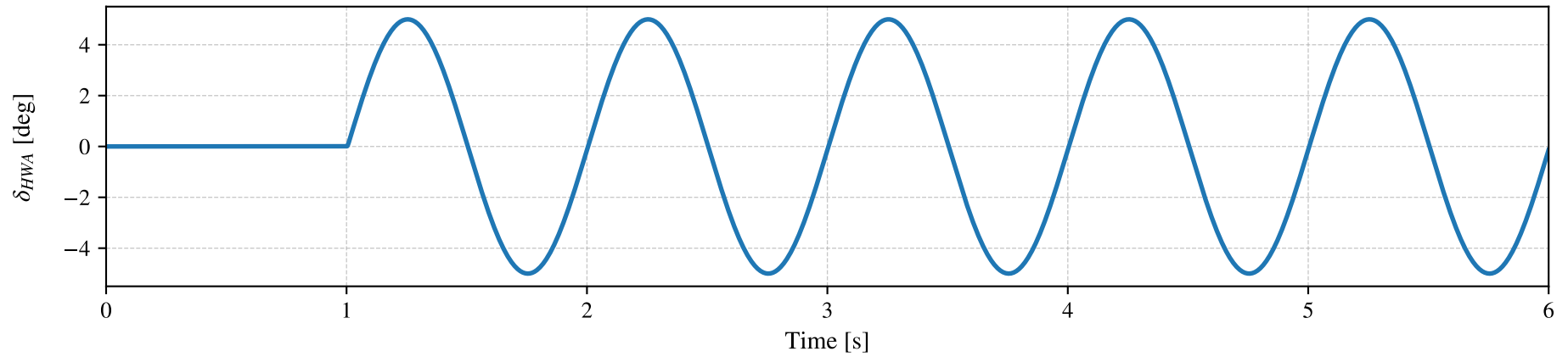


One-Period Sinusoidal Response (TransientEval)

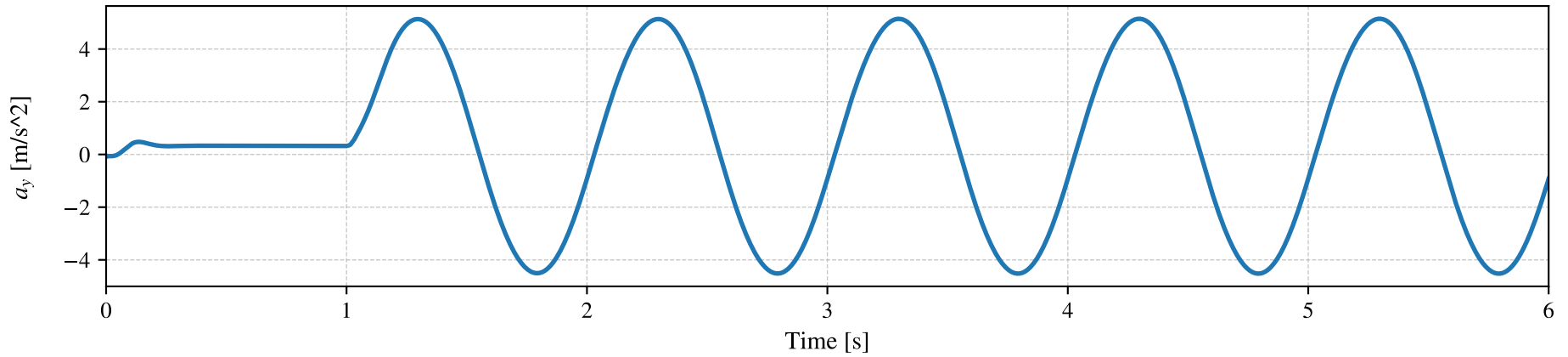


Continuous Sinusoidal Response

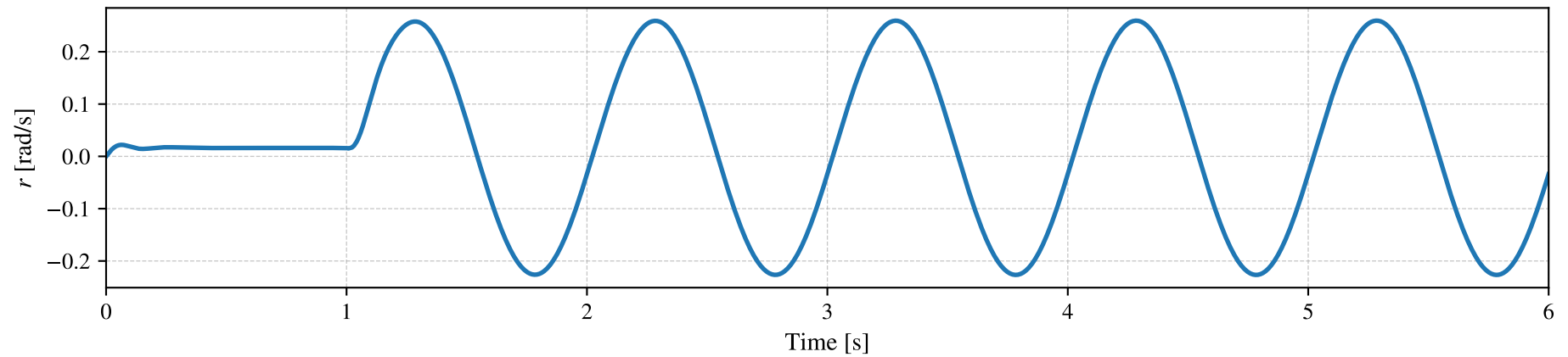
Steering Wheel Angle δ_H



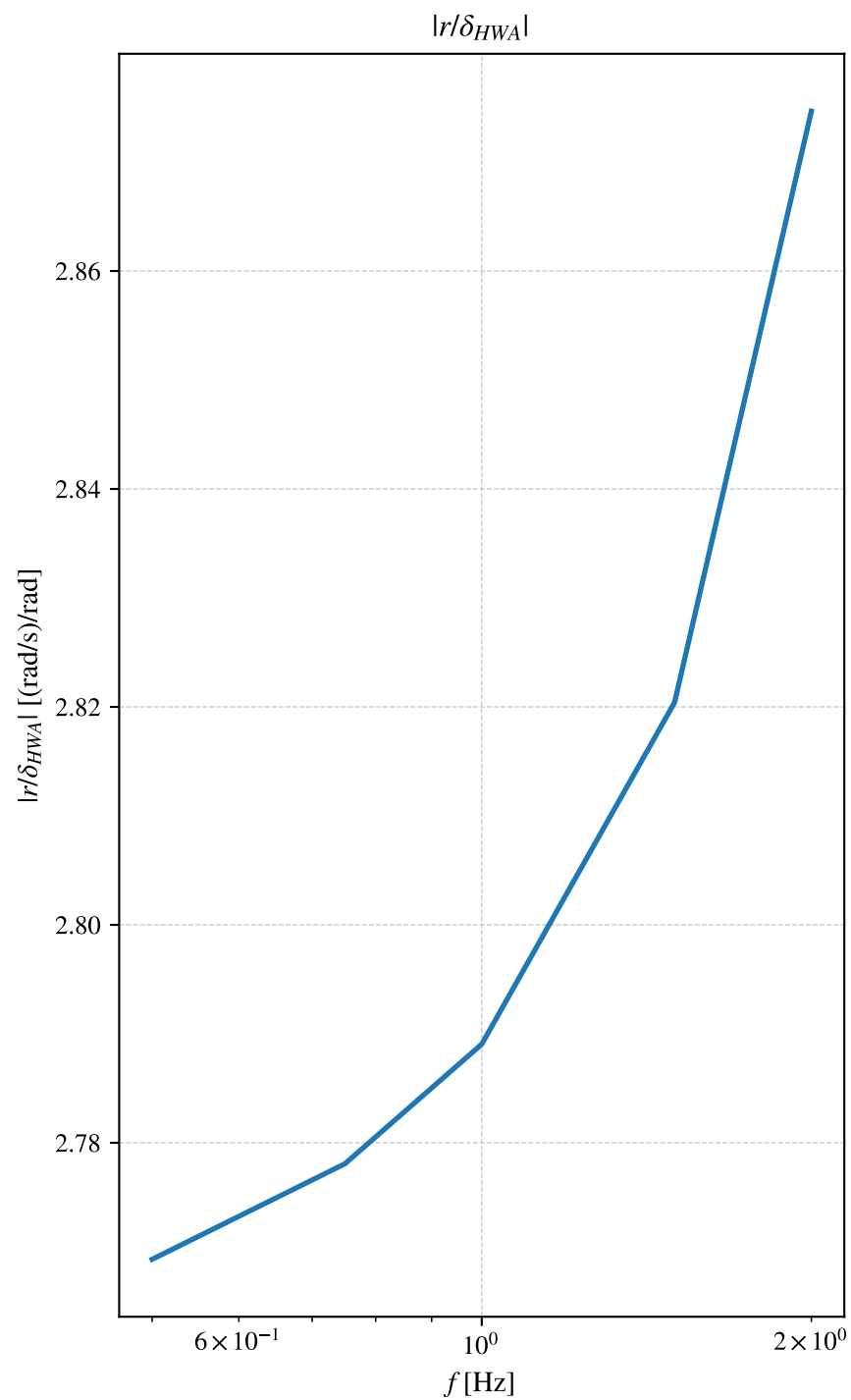
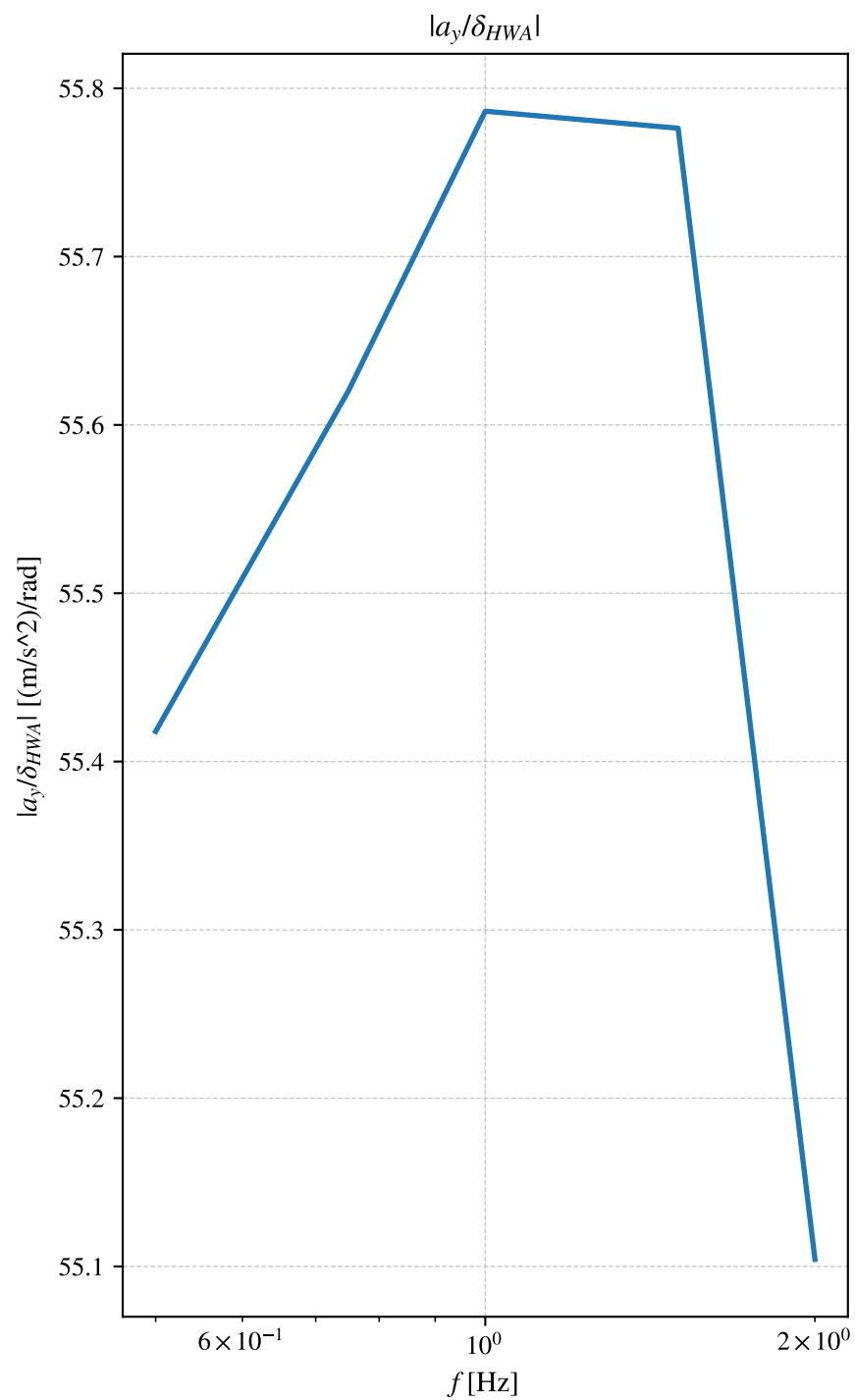
Lateral Acceleration a_y



Yaw Velocity r

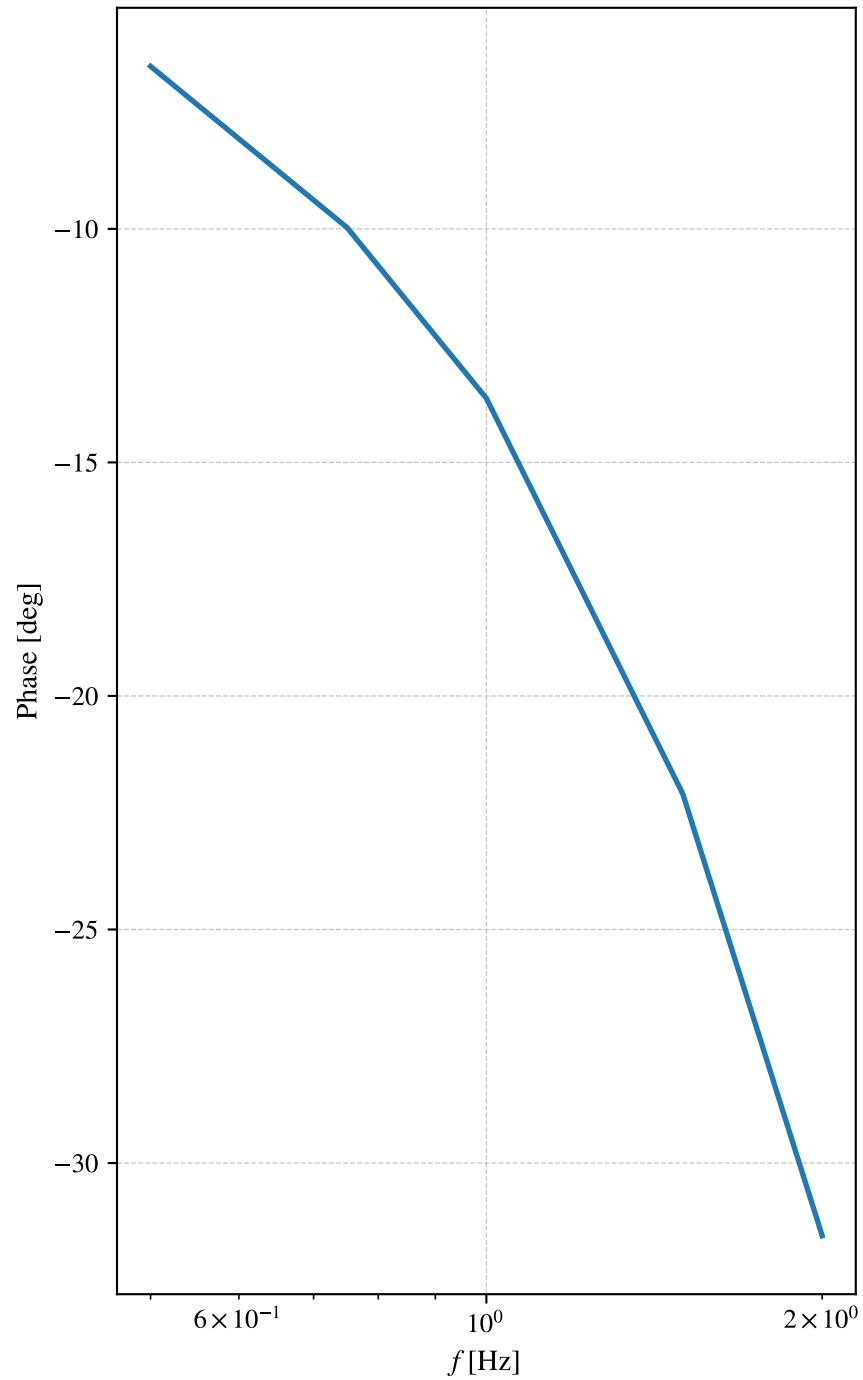


Frequency Response — Gain (TransientEval)



Frequency Response — Phase (TransientEval)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

